Summary of the Ram Calculus Exponential-Time-Hypothesis Report

Contributions Relative to the OpenAI Paper,
Proved Refinements, Matched Benchmarks, and Open Research Directions

Revision 2

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Technical Report Prepared with the Assistance of AI Models
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October 26, 2026

Abstract

This report summarizes the completed investigation *Ram Calculus and the Exponential Time Hypothesis: Exact Finite-Algebra Reformulations, Aggregated Boolean Jets, and Precision-Certified Counting*. It explains which results are inherited from OpenAI's October 25, 2026 manuscript, which are equivalent Ram-compatible reformulations, and which are additional proved refinements. The most concrete algorithmic addition replaces a subset-pair loop by a subset loop with a small exact generating state, reducing restricted-count calls while increasing coefficient arithmetic. A weighted-support extension applies under certified higher-order vanishing. Exact row selection preserves counting fibers, and target-specific two-adic Hermite certificates permit smaller precision budgets in two tested finite configurations. The summary includes the key formulas, proof mechanisms, worked examples, experimental scope, and limitations. Revision 2 adds matched count, runtime, and memory tests on 21 certified finite layouts, including five with node-varying factor assignments. All exact outputs agree; positive-order cases show lower aggregate medians in these runs, but the extra coefficient state and zero-order overhead remain visible. No auxiliary PDE was found in the source paper, and no quaternionic speedup is established. The report does not prove a stronger general SAT exponent, an independent complete refutation of ETH, or optimality of the source bound. Its defensible contribution is a set of exact local refinements and an explicit framework for identifying what must still be implemented, bounded, and verified.

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1 Purpose and principal conclusions

The document being summarized is the revised technical report *Ram Calculus and the Exponential Time Hypothesis: Exact Finite-Algebra Reformulations, Aggregated Boolean Jets, and Precision-Certified Counting* [1], called **the Ram report** below. The primary source is OpenAI's 66-page manuscript *The Exponential Time Hypothesis is False*, dated October 17 and revised October 25, 2026 [2], called **the source paper**. This summary explains the completed investigation; it does not claim a new literature survey, fresh large-instance experiments, or a further complete proof audit.

The principal conclusions are:

1. **There is substantive relevance, but in finite algebra rather than PDE inversion.** Exact moments, finite polynomial operators, jets, aggregate states, transported products, and residual certificates match specific steps of the source algorithm.
2. **A local call-count improvement is proved.** The Ram report's Theorem 8.1 and Corollary 8.2 replace $\sum_{j\le q} 2^j\binom{n}{j}$ restricted calls by $\sum_{j\le q}\binom{n}{j}$ calls, with polynomially enlarged coefficient states under the stated size assumptions.
3. **Additional structural and precision refinements are proved.** Weighted support budgets, exact redundant-row removal, and target-specific two-adic reconstruction bounds have explicit hypotheses and proofs.
4. **Some improvements are demonstrated only in finite tests.** Strict precision savings occur in two small Hermite configurations. The original tests check identities and certificates; revision 2 additionally benchmarks finite matched implementations. Neither is a benchmark of the complete claimed subexponential SAT algorithm.
5. **The headline complexity claim is not strengthened.** The report gives neither a better unrestricted SAT exponent nor a new independent end-to-end proof of the source's theorem. Its local refinements are also expressible using conventional algebra, so no uniquely Ram-exclusive speed advantage has been established.

Throughout this summary, *new contribution* means an addition supplied with a proof or computation in the Ram report relative to its treatment of the source paper. It is not a verified assertion of historical priority. Finite differences, generating functions, Hermite interpolation, quotient algebras, and valuation estimates have classical antecedents.

2 What the OpenAI paper claims and what it contributes

2.1 The exact problem and the complexity benchmark

For a CNF formula F on n declared Boolean variables and m clauses of width at most a fixed integer k, the exact counting target is

$$Z(\mathsf{F}) = \sum_{x\in\{0,1\}^n} \mathbf{1}_{\{\mathsf{F}(x)=1\}}. \tag{1}$$

It is an integer between zero and 2^n. The decision problem asks whether it is positive. Unused declared variables still contribute factors of two. Thus normalization, parity, and exact multiplicity matter: a probability, a count modulo two, and the integer count are different observations.

ETH is the hypothesis that, in its standard bounded-clause-width setting, 3-SAT has no subexponential-time algorithm. The source claims deterministic exact counting in the following

bit-time bounds for fixed parameters:

$$T(n,m) = (n+m)^{O_k(1)} 2^{O_k(n \log\log n/\sqrt{\log n})}, \tag{2}$$

$$T(n,m) = (n+m)^{O_{k,D}(1)} 2^{O_{k,D}(n/\sqrt{\log n})}, \quad m \le Dn, \tag{3}$$

with polynomial bit space. Its earlier interpolation route has the weaker scale

$$(n+m)^{O_k(1)} 2^{O_k(n(\log\log n)^2/\sqrt{\log n})}. \tag{4}$$

These are the *source's* bounds as recorded and analyzed in the Ram report, not new bounds obtained by applying Ram Calculus. For fixed k, D the exponents are $o(n)$, but this does not imply polynomial time, practical finite-size performance, or uniform constants when the parameters grow.

2.2 The difficult ingredients belong to the source paper

The source does much more than express a count as a polynomial. Its construction deforms variable-incidence rows into a polynomial family that restores the original variables at parameter zero and has special separation properties at many other nodes. Bounded integer summaries then mediate the coupling between groups of variables. Boolean finite differences recover short jets, and exact interpolation recovers the desired count. The reductions use parity checks, modulus-amplifying indicators, and degree control. The sharper Appendix D route uses Boolean tensor networks, Cauchy deformation, scaled jets, valuation-sensitive mode truncation, and recursive contraction.

The challenging point is to obtain *all* the required properties simultaneously: small enough deformation degree, enough useful nodes, small summary ranges, manageable coefficients, exact arithmetic, and controlled live storage. A generic transform or finite inverse does not provide that construction. The Ram report retains these source ingredients rather than replacing them with an unproved universal inversion claim.

2.3 The level of verification must remain explicit

The Ram report used the supplied PDF and matching TeX at commit `42907c9e999cc00d6abe7a50bfc1023250edd1b9`. Its separate audit did not find a demonstrated error in the examined identities, construction, arithmetic accounting, and Appendix D arguments. That is a bounded audit, not a global correctness certificate. Appendix A's full combinatorial proof, all external prerequisites, and circuit consequences were not independently re-proved.

The inspected Lean interface contains an exact-counting and SAT statement at the weaker scale (4). The comparator specification's intentional placeholder is not evidence of a gap in the separate solution module. Conversely, inspecting those files does not establish a clean build, an axiom audit, correspondence of every encoding with the paper, or formal verification of the sharper Appendix D bounds. The Ram investigation did not execute that full dependency closure. Its local theorems should therefore be evaluated on their own explicit proofs.

3 Where Ram Calculus enters, and where it does not

3.1 An exact representation must retain the whole algebra

On the Boolean cube, functions have unique multilinear representatives in

$$\mathcal{B} = \mathbb{Q}[X_1, \dots, X_n]/(X_i^2 - X_i : 1 \le i \le n).$$

For $f = \sum_{S\subseteq[n]} c_S X_S$, where $X_S = \prod_{i\in S} X_i$, the exact sum functional is

$$\Lambda(f) = \sum_{S\subseteq[n]} 2^{n-|S|} c_S. \tag{5}$$

Each monomial equals one on $2^{n-|S|}$ assignments. This elementary observation connects counting to Ram moment coordinates, but the algebra still has dimension 2^n.

If an invertible coordinate map R is used, both multiplication and the observation must be transported:

$$u \star v = R((R^{-1}u)(R^{-1}v)), \qquad \Lambda_R(u) = \Lambda(R^{-1}u). \tag{6}$$

Ordinary componentwise multiplication in arbitrary new coordinates would usually solve a different problem. These formulas are exact rederivations, not computational compression theorems.

3.2 The concrete corpus contributions

The Ram report's corpus survey was targeted, not an assertion that every page in the large corpus was audited. Its strongest connections are:

- **RamOperator Foundations and the revised Ram Series report:** finite polynomial operators, terminating inverses on appropriate finite spaces, dual observation rows, and projection-leakage safeguards.
- **TR-item-010 and corrected TR166:** Boolean quotients, conditional closure, rank-versus-inversion distinctions, and explicit conditions for exact recovery.
- **OpenAI-722 reports M012 and M231:** finite-alphabet moments, dimension accounting, characteristic-safe jets, and structured arithmetic.
- **Revised foundational Volume 1:** Newton and divided-difference coordinates with the correct product and observation rules.
- **RMT extensions TR1–TR4:** exact aggregate states, multiplicity and jet budgets, ordered hypercomplex channels, and nonlinear range-consistency constraints. Their limitations are as important as their useful interfaces.

The scalar finite-algebra results are more directly relevant here than continuous PDE or quaternionic physical-field reconstruction. The central new identities are self-contained finite-sum and valuation arguments; they do not depend on treating the entire confidential corpus as axioms.

3.3 Exact shared states: a rederivation, not a newly invented factorization

Partition the variables into groups S_g, and retain integer summaries $y_g = C_g x_g$. Suppose each grouped factor can be written as $a_g(x_g, y_{-g})$. Then, with any compatible fixed-bit restrictions,

$$\sum_x \prod_g a_g(x_g, y_{-g}) = \sum_y \prod_g \left(\sum_{x_g : C_g x_g = y_g} a_g(x_g, y_{-g}) \right). \tag{7}$$

This follows because every assignment lies in one fiber and, once y is fixed, that fiber is a product of local fibers. The source already uses this separation. Ram aggregate-state language emphasizes its certificate: all outside dependence must pass through the retained state, and all assignments with

that state contribute their weights. Keeping only a minimum-cost representative, appropriate for some optimization problems, would not preserve the count.

For polynomial-deformation rows, the off-group dependence condition is an exact row-space relation $P_{f,v}(t)|_{S_h} \in \mathrm{span}_{\mathbb{Q}}(\text{rows of } C_h)$. This can be checked by rational linear algebra. Low rank alone is not enough for efficiency: summary magnitudes and attainable values must also be bounded.

4 Main proved addition: aggregating the Boolean-jet loop

4.1 Why the source has a repeated subset loop

Let $H(T)$ be the source's deformed counting polynomial, with $H(0) = Z$ and $\deg H \le dR$, where d bounds the incidence-separated product degree and R bounds the deformation-row degree. At a useful node t, introduce independent Boolean placeholders X and the formal increment z:

$$G_x(X, z) = \prod_f \Phi_f\big((P_{f,v}(t) \cdot x + (P_{f,v}(t+z) - P_{f,v}(t)) \cdot X)_{v \in I_f}\big).$$

Each occurrence of X_i costs at least one power of z, hence $\deg_X [z^j] G_x(X, z) \le j$. Also $\sum_x G_x(x, z) = H(t+z)$. Fixing X leaves the source's separation in x available; immediately substituting $X = x$ can destroy it.

For Q nodes, take $q = \lfloor dR/Q \rfloor$ and $J = q + 1$. Then $QJ > dR$, so J coefficients per node determine the polynomial. For a family of polynomials $p_x(X)$ of degree at most q, Boolean subset inversion gives

$$\sum_x p_x(x) = \sum_{\substack{E \subseteq [n] \\ |E| \le q}} \sum_{U \subseteq E} (-1)^{|E|-|U|} \sum_{x_E = 1} p_x(\mathbf{1}_U). \tag{8}$$

Here $x_E = 1$ means every coordinate indexed by E is one, and $\mathbf{1}_U$ is the indicator vector of U. The source's corresponding restricted-call count per node is

$$C_q(n) = \sum_{j=0}^{\min(q,n)} 2^j \binom{n}{j}. \tag{9}$$

The extra 2^j comes from separately enumerating every $U \subseteq E$.

4.2 The new identity and its short proof

Define the integer weight

$$w_q(h, u) = \sum_{k=0}^{\min(q-u, h-u)} (-1)^k \binom{h-u}{k}, \qquad h \ge u, \quad u \le q. \tag{10}$$

The Ram report's Theorem 8.1 proves

$$\boxed{\sum_x p_x(x) = \sum_{\substack{U \subseteq [n] \\ |U| \le q}} \sum_{x_U = 1} w_q(|x|, |U|) p_x(\mathbf{1}_U).} \tag{11}$$

The symbol $|x|$ denotes Hamming weight. To see the identity, interchange the finite sums in (8). For fixed x and $U \subseteq \mathrm{supp}\, x$, all admissible E have the form $E = U \cup K$, where $K \subseteq \mathrm{supp}\, x \setminus U$ and

$|K| \le q - |U|$. There are $\binom{|x|-|U|}{k}$ possibilities with $|K| = k$, each with sign $(-1)^k$. Summing those possibilities gives exactly (10).

Thus the added result is not a different answer or an approximation. It is an exact aggregation of repeated work in this particular source routine. Its proof does not assume the full ETH theorem.

4.3 Why the Hamming weight does not break the source factorization

The global-looking weight can be implemented with a small generating state. For fixed U, let $r_U = q - |U|$ and compute

$$K_U(z, w) = \sum_{x_U=1} G_x(\mathbf{1}_U, z)(1 + w)^{|x|-|U|} \pmod{z^J, w^{r_U+1}}. \tag{12}$$

Since

$$(1 + w)^{|x|-|U|} = \prod_g (1 + w)^{|x_g|-|U\cap S_g|},$$

the new factor is local to each group. The original measurement summaries still suffice. After the group products are completed, apply the linear functional

$$\mathcal{L}_{r_U}\left(\sum_{k=0}^{r_U} b_k w^k\right) = \sum_{k=0}^{r_U} (-1)^k b_k.$$

Then $H(t + z) = \sum_{|U|\le q} \mathcal{L}_{r_U} K_U(z, w)$ modulo z^J.

This is the essential Ram-style design choice: retain just enough exact coefficient state to summarize the eliminated combinatorial index. The construction is also a conventional generating-function calculation. It does not use an unproved compression of arbitrary Boolean assignments.

A necessary order-of-operations restriction. The final functional is not a homomorphism of the truncated ring. Modulo w^2, the product $w \cdot w$ is zero, but evaluating each retained factor at -1 first and multiplying gives one. Therefore the final functional must follow multiplication; moving it inside products gives wrong counts.

4.4 What improves and what must still be paid for

There is now one restricted call per U, so Corollary 8.2 gives

$$B_q(n) = \sum_{j=0}^{\min(q,n)} \binom{n}{j}, \qquad B_q(n) \le C_q(n) \le 2^{\min(q,n)} B_q(n). \tag{13}$$

An element of the enlarged coefficient algebra needs at most $J(q + 1)$ base-algebra slots. Naive multiplication uses $O(J^2(q+1)^2)$ base-algebra operations. Polynomial coefficient bit sizes, polynomial q, J, and a streamed implementation are necessary to preserve the claimed polynomial workspace per examined assignment.

With the source's group-size and measurement bounds, the corresponding assignment-examination estimate changes from

$$2Qr\, C_q(n)(2D + 1)^{2\tau n} 2^{2n/r} \quad \text{to} \quad 2Qr\, B_q(n)(2D + 1)^{2\tau n} 2^{2n/r}.$$

n	q	Source loop C_q	Aggregated loop B_q
20	2	801	211
40	4	1,544,481	102,091
80	8	7,847,456,478,273	32,490,460,747
6	6	729	64

Table 1: Exact combinatorial call counts from the Ram report. These numbers were calculated from binomial formulas; the large loops were not executed.

Factor evaluation and bit arithmetic are additional costs. Consequently, **fewer restricted calls is proved; a faster complete solver is not.**

For $q = o(n)$, both combinatorial logarithms have leading form $q \log(n/q) + O(q)$. The summary-range and group-enumeration costs also remain. Moreover, this refinement concerns the source's earlier Section 3 route, not a demonstrated modification of its sharper Appendix D recursion. It does not yield a new bound replacing (2) or (3). When $q \geq n$, it reduces 3^n calls to 2^n but then selects direct Boolean enumeration; that regime is not subexponential.

4.5 A small worked example

For $n = 3$, $q = 1$, let $p_x(X) = a_x + \sum_i b_{x,i} X_i$. The empty-set weight is $1 - |x|$, while every allowed singleton weight is one. For fixed x, the aggregated contribution is

$$(1 - |x|)a_x + \sum_{i : x_i = 1} (a_x + b_{x,i}) = a_x + \sum_i b_{x,i} x_i = p_x(x).$$

The original loop has $1 + 2n = 7$ calls and the new one $1 + n = 4$. The cancellation of repeated constant terms makes the improvement transparent. It also illustrates why signed coefficients must be retained.

5 Other proved structural refinements and their conditions

5.1 Weighted support budgets

Suppose coordinate i first occurs in a correction at order z^{ν_i}, with a *proved* integer $\nu_i \geq 1$. Then a Boolean support E costs at least $\nu(E) = \sum_{i \in E} \nu_i$ powers of z. The Ram report's Theorem 9.1 replaces the ordinary size constraint by the downward-closed family $\{E : \nu(E) \leq q\}$. Its aggregated identity is

$$\sum_x p_x(x) = \sum_{\nu(U) \leq q} \sum_{x_U = 1} p_x(\mathbf{1}_U) \sum_{\substack{K \subseteq \operatorname{supp} x \setminus U \\ \nu(K) \leq q - \nu(U)}} (-1)^{|K|}, \tag{14}$$

provided each Boolean normal form has support only in that family. The proof is the same finite-sum rearrangement, now with a weighted budget. The inner weight is the sum of the retained coefficients of

$$\prod_{i \in \operatorname{supp} x \setminus U} (1 - s^{\nu_i}) \pmod{s^{q - \nu(U) + 1}},$$

which again factors locally.

This yields $|\{U : \nu(U) \le q\}|$ restricted calls. With $(\nu_1, \nu_2, \nu_3) = (1, 2, 3)$ and $q = 2$, the allowed sets are $\varnothing, \{1\}, \{2\}$: three calls instead of the seven unweighted subset calls. The theorem is exact, but its usefulness for the source algorithm is conditional. A construction with many higher-order zeros may require greater deformation degree R, larger summaries, or more expensive coefficients. Those costs can cancel the saving. The report does not prove that a better globally balanced construction exists.

5.2 Exact redundant-row removal

If C' consists of a subset of rows of an integer summary matrix C spanning the same rational row space, then

$$Cx = Cx' \quad \Longleftrightarrow \quad C'x = C'x'. \tag{15}$$

Both matrices have the same kernel; each omitted row is a rational linear combination of retained rows. Therefore the fibers, their multiplicities, and the off-group row-space certificates are unchanged. Selecting existing rows avoids magnitude growth from an arbitrary rational basis change.

This can tighten a conservative summary box to $\prod_j (2D_j + 1)$ using retained row norms D_j. It is a proved, instance-dependent preprocessing refinement, using standard exact linear algebra. It gives no strict saving when the original rows are already independent. Nor does rank bound the number of values: the rank-one row $(1, 2, 4, \ldots, 2^{n-1})$ has 2^n distinct Boolean measurements.

5.3 Restricted-model results are not general SAT improvements

The report rederives standard exact counting on a supplied tree decomposition of primal-graph width w in $2^{w+1}\,\mathrm{poly}(n, m, |\mathcal{T}|)$ bit operations. Messages count extensions conditional on separators, preserving multiplicities. This is a useful independent exact-closure example, not a new general algorithm: bounded clause width does not guarantee small treewidth, and finding a decomposition may itself cost time.

It also records a rank formula over the Boolean function algebra over $\mathbb{F}_2$. If M_i is multiplication by a reduced constraint polynomial P_i, then the integer number of common zeros is

$$N = 2^n - \mathrm{rank}_{\mathbb{F}_2}[M_1 \ \cdots \ M_m].$$

Evaluation diagonalizes the Boolean algebra and proves the identity. Its matrices nevertheless have 2^n rows. This is an exact reference representation, not a subexponential rank algorithm. These examples separate correctness of a reformulation from the cost of using it.

6 Second main addition: target-specific precision certificates

6.1 What is already in the source, and what is added

The sharper source route reconstructs a polynomial's value at zero from scaled Hermite jets. It already uses a target-at-zero formula and a conservative precision allowance. The Ram report does *not* claim to invent targeted Hermite interpolation. Its addition is to calculate the actual reconstruction weights and use their two-adic valuations as coordinatewise certificates, rather than only using a uniform upper envelope.

Let F be a polynomial, t distinct rational nodes, q_t their multiplicities, and $N = \sum_t q_t > \deg F$. For a nonzero rational scaling α, define $h_{t,j} = [z^j]F(t + \alpha z)$ for $0 \le j < q_t$. There are unique weights

for the full degree-less-than-N Hermite reconstruction map such that

$$F(0) = \sum_t \sum_{j<q_t} \lambda_{t,j} h_{t,j}. \tag{16}$$

They can be obtained by exact rational elimination from $A^T\lambda = e_0$, where $A_{(t,j),k} = \binom{k}{j} t^{k-j}\alpha^j$ for $k \geq j$ and zero otherwise. An equivalent cardinal-polynomial formula is provided in the full report.

6.2 The proved certificate and why it works

Write $v_2(r)$ for the exponent of two in a nonzero rational number r. Negative values identify powers of two in a denominator. Suppose the target is integral at two and the jet errors satisfy $v_2(\widehat{h}_{t,j} - h_{t,j}) \geq b_{t,j}$. Theorem 13.1 proves that

$$b_{t,j} + v_2(\lambda_{t,j}) \geq p \quad \text{for every nonzero weight} \tag{17}$$

is sufficient for the reconstructed value to be correct modulo 2^p. Indeed, every weighted error is divisible by 2^p in the two-local coefficient module; their sum is too. This uses the valuation inequality $v_2(a+b) \geq \min(v_2(a), v_2(b))$.

A sufficient shared precision for all coefficients at node t is

$$p_t = p + \ell_t, \qquad \ell_t = \max\left(0, \max_{j:\lambda_{t,j}\neq 0} -v_2(\lambda_{t,j})\right). \tag{18}$$

Zero weights require no accuracy for this target. If the recursive interface requires a strict precision increase, use $p + \max(1, \ell_t)$. The fixed-lift convention of the source recursion must still be preserved; this is not permission to lower arbitrary intermediate precisions independently.

For fixed nonzero weights and independently variable errors, the coordinatewise condition is sharp: a single error one valuation level below its threshold produces a target error of valuation $p-1$. This is *not* a universal optimality theorem. A smaller polynomial space could admit other recovery weights, and correlated errors could allow further cancellations. The proved statement is about the declared linear recovery map and error model.

6.3 Strict finite savings actually demonstrated

The tests computed exact rational weights for two small tapered grids. Table 2 lists extra precision bits above the desired output precision p. For the first grid, both multiplicities are three and $\alpha = 4$.

Configuration and node class	Source allowance	Actual certified loss
$L = 2$, nodes $-1, 1$	64	7
$L = 4$, odd nodes $-3, -1, 1, 3$	144	33
$L = 4$, even nodes $-2, 2$	96	7

Table 2: Local exact precision-certificate improvements. The last two rows belong to the same grid, not two additional independent configurations.

For the second, odd-node multiplicities are six, even-node multiplicities are three, and $\alpha = 8$. These configurations meet the local interpolation hypotheses but not the large-size packing assumptions of

the complete source recursion. Ten additional polynomial reconstructions agreed with the exact value at zero.

The savings are genuine reductions of a sufficient local certificate: for example, the first configuration needs only $p+7$ rather than the conservative $p+64$ bits. They are not measurements of full-algorithm runtime, memory, or a changed asymptotic exponent. Computing the weights also has a cost in their number and bit lengths.

The general estimate in the report remains of the same order as the source envelope. In its tapered notation it proves $\ell_t \le 6hD_t+4M(a+1)$, bounded by the source allowance $8hD_t+8M(a+1)$. Better constants and finite cancellations do not by themselves establish a better uniform growth rate for the recursion.

6.4 Ordinary numerical accuracy is a different question

For absolute real or complex jet errors $|e_{t,j}| \le \delta_{t,j}$, the relevant certificate is

$$|\widehat{F}(0) - F(0)| \le \sum_{t,j} |\lambda_{t,j}|\delta_{t,j}. \tag{19}$$

If the target is an integer and this bound is strictly below $1/2$, rounding recovers it exactly. For a normalized expectation equal to $2^{-n}Z$, the expectation error must be below 2^{-n-1} to guarantee correct count rounding. Small unweighted residuals or ordinary relative accuracy are insufficient. This is why exact modular computation cannot simply be replaced by an approximate Ram transform with no target-error certificate.

7 What the tests establish, and what they do not

The full report's program uses Python integers and `fractions.Fraction`, with seed `20261026`. The existing files `eth-support/exact_identity_tests.py` and `eth-support/test-results.json` accompany the main report. The following are the reported completed checks, not a newly executed large-scale evaluation in this summary.

Table 3: Exact tests recorded in the Ram report; all passed.

Check	Cases	Meaning
Boolean rational identities	132	Direct sums, the original pair loop, the aggregated identity, and generating jets agree for $n = 1, \dots, 6$ and $q = 0, \dots, n+1$.
Weighted-support identities	120	Weighted aggregation agrees with direct sums for weights in $\{1, 2, 3\}$ and varied budgets.
Shared-fiber generating jets	24	Two three-bit groups with cross-group summary dependence give matching direct and factored coefficient vectors.
Rational Hermite recovery	36	Unequal-multiplicity reconstruction agrees exactly for degrees one through nine.
Parity lifts and shared-order truncation	360	Modular indicators and truncated products agree modulo 2^ℓ for $\ell = 1, \dots, 8$.

Check	Cases	Meaning
Scaled Cauchy-integrality rows	1,168	The checked grids $L = 2, 4, 8$ have the required odd-denominator scaled coefficients, origin identity, and cross-group divisibility.
Tapered Hermite configurations	2	Actual weight valuations give Table 2; ten polynomial reconstructions also agree.
Negative controls	4	Deliberately invalid shortcuts are detected.

Finite tests corroborate the implementations of the stated identities; the general validity comes from the proofs with their assumptions. They do not establish large-n parameter inequalities, practical crossover sizes, end-to-end bit complexity, all source prerequisites, or correctness of a Lean build that was not run. The additional matched implementation evidence is reported separately in Section 9; it does not replace these identity tests.

7.1 A transparent SAT calculation

For $\mathsf{F} = (x_1 \vee x_2) \wedge (\neg x_1 \vee x_3)$, the satisfaction polynomial is $(x_1 + x_2 - x_1x_2)(1 - x_1 + x_1x_3)$. Replacing each variable by $(1 - T)x_i$ and summing over all eight assignments produces

$$H(T) = 4 - 3T + T^2 - T^3 - T^4, \qquad H(1 + z) = -8z - 8z^2 - 5z^3 - z^4.$$

Thus $H(0) = 4$, and the expansion around one recovers $8 - 8 + 5 - 1 = 4$. But $H(2) = -22$: intermediate deformed values need not be nonnegative counts. Signed exact arithmetic is essential. This example checks the algebra and does not satisfy the asymptotically advantageous low-degree design conditions; it is not a fast SAT solver.

7.2 Failure cases rule out tempting shortcuts

1. **Early final-functional application.** Evaluation of a retained polynomial at ± 1 is generally not multiplicative after truncation. The zero product w^2 modulo w^2 provides an immediate counterexample.

2. **Rank-only state compression.** One binary-weight measurement on eight bits has 256 distinct values despite rank one.

3. **Uncertified floating cancellation.** Exact integers give $(2^{60} + 1) - 2^{60} = 1$, whereas binary64 arithmetic gives zero.

4. **Differentiation inside a truncated algebra.** Ordinary differentiation does not descend as a derivation of $\mathbb{Q}[z]/(z^J)$. For $J = 2$, a purported Leibniz rule would differentiate the zero product z^2 to the nonzero element $2z$.

These controls reject invalid alternative implementations. They are *not* errors found in the OpenAI paper. Likewise, identical low-order moments need not identify the full Boolean distribution: even- and odd-parity distributions have matching proper-coordinate marginals and different supports.

8 PDEs, quaternionic Ram Calculus, and the scope of TR3

8.1 No auxiliary PDE was identified in this source

The auxiliaries in the ETH manuscript are finite polynomials, formal order variables, subset differences, integer summaries, and ring-valued tensor networks. They are not an elliptic, parabolic, or hyperbolic boundary-value problem whose solution yields the count.

It is possible to define $u(t, z) = H(t + z)$ and write $\partial_t u = \partial_z u$. This is a formal translation identity in characteristic zero. Its initial coefficients already encode the hard count, so it gives neither missing information nor a cheaper algorithm. On a bounded-degree polynomial vector space, translation is the finite operator exponential

$$p(t + z) = \sum_{j=0}^{D} \frac{z^j}{j!} p^{(j)}(t).$$

That legitimate Ram-operator connection must not be confused with solving a new PDE or differentiating a truncated quotient using an invalid rule. The auxiliary Dirichlet equation in the earlier local-smoothing paper is a different construction and cannot be transferred to ETH by analogy.

8.2 The useful role of corrected hypercomplex theory is caution

The source's grouped products use a commutative associative coefficient algebra. Corrected quaternionic TR60 and RMT extension TR3 specify ordered left/right actions and physical metrics. They do not permit unrestricted reordering: quaternion units satisfy $ij = k$ and $ji = -k$. Real block representations preserve an ordered linear action, not commutativity or automatically smaller tensor-network width. Octonions also require parenthesization because associativity fails.

Consequently, the report establishes no counting improvement from these hypercomplex extensions. Their value here is to prevent an invalid substitution and to delimit possible future vector-valued variants. Similarly, TR4's nonlinear range-consistency and local residual certificates do not enumerate every Boolean solution. Local inversion and exhaustive exact counting are different mathematical obligations.

9 Revision 2: matched implementation results

The earlier report established identities and counted calls. The revised main report additionally implements both exact pipelines and tests them on 21 matched CNFs. Sixteen have fixed two-group layouts; five change factor assignment between nodes and do not permit the same direct-at-zero fiber shortcut. These are certified finite Section 3 inputs, not instances generated by the source's Section 4 asymptotic construction. The comparison is against an independent Python implementation of the source formulas, not an OpenAI executable or the stronger Appendix D algorithm.

For each case both methods use the same clauses, rows, measurements, nodes, truncation, exact arithmetic, streamed scans, and full Hermite reconstruction. The original retains all (E, U) restricted calls, with fixed-U corrections hoisted for both methods. The aggregate replaces the pair loop by a U loop with a truncated generating state. No other proposed Ram optimization is mixed into the comparison.

Every exact count, jet coefficient, and reconstructed polynomial coefficient matched exhaustive independent oracles in all 294 timed executions and the separate memory and diagnostic runs. The deformation oracle uses independent convolution and enumeration but shares row construction and its dot-product helper; the explicit layout proofs and static audit separately check that shared part. Each method has seven fresh-worker repetitions per case, with randomized order within each pair. The pipeline timer includes preparation, jet computation, and recovery, but not process launch, imports, input reading, oracle work, or serialization. The active Windows laptop was not pinned or load-controlled, so the recorded ranges matter.

Table 4: Matched median pipeline times (milliseconds), with the ratio original/aggregated. Complete raw ranges and phase data appear in the main report and accompanying JSON files.

ID	q	Exact count	Original ms	Aggregate ms	Ratio
S01	0	16	0.466	0.539	0.87
S02	0	0	1.069	1.135	0.94
S03	0	32	3.804	3.779	1.01
S04	1	10	4.843	3.756	1.29
S05	1	30	18.564	12.461	1.49
S06	1	96	48.059	33.333	1.44
S07	2	50	196.037	70.523	2.78
S08	2	192	250.717	86.236	2.91
S09	2	784	1,889.863	546.129	3.46
S10	3	48	398.996	119.954	3.33
S11	3	168	3,042.474	607.451	5.01
S12	2	50	907.584	354.187	2.56
S13	3	704	9,833.618	1,643.942	5.98
S14	4	168	3,895.762	291.173	13.38
S15	6	8	140.890	38.303	3.68
S16	2	24	1,987.977	655.549	3.03
N17	2	8	50.535	24.907	2.03
N18	2	40	208.674	131.844	1.58
N19	3	52	880.379	225.510	3.90
N20	3	192	3,488.846	501.599	6.96
N21	4	44	2,667.340	473.529	5.63

Every one of the 18 cases with $q > 0$ had a shorter aggregate median in these runs, with median ratios from 1.29 to 13.38. The largest ratio occurs at S14; it is descriptive and not a forecast of scaling. The two smallest controls S01 and S02 were slower with aggregation, while S03 was nearly tied. These three have $q = 0$ and exactly the same number of restricted calls. The wide raw ranges, particularly on the busier runs, preclude a strong claim based on the precise ratio digits. Exact call and assignment-visit reductions corroborate the mechanism independently of wall-time noise.

Memory is a tradeoff, not a uniform gain. Whole-pipeline traced peak ratios A/O range from 0.991 to 1.033; there is no uniform memory reduction. Common layout and dense Hermite-recovery allocations can dominate the peak. For $q > 0$, observed kernel incremental-peak ratios range from 0.98 to 1.86. These are net phase measurements, not a count of just the extra generating coefficients; released buffers and stored output jets affect them. The logical coefficient capacity per ring element grows from J to at most $J(q + 1)$, so a speed benefit is compatible with a larger kernel memory footprint. Untraced timing workers had lifetime process peaks between 24.56 and 25.24

MiB, including interpreter and imports. Traced-worker process peaks, retained in the raw data, also contain tracer overhead; neither process measure is an isolated solver-workspace bound.

A stronger structural control prevents overstatement. The first 16 fixtures already separate at $T = 0$, so direct fiber counting matches their counts without any jet recovery. The direct-control medians range from 0.145 to 4.390 ms (all individual samples: 0.139–9.446 ms). They are shorter than the corresponding aggregate medians in every first-suite fixture, although these later, unpaired measurements are not suitable for a precise speedup estimate. The control even includes unnecessary Hermite-matrix preparation, so it is not a fully optimized direct implementation. That shortcut is not valid for the five node-varying cases and was not applied to them. They therefore test a less trivial use of the jet mechanism, while still involving at most eight variables and no proof of a new general complexity bound.

What is newly demonstrated. The revision supplies measured finite runtime gains, actual coefficient and workspace costs, and failure/overhead controls for a reproducible implementation. It does not establish that aggregation is faster for every implementation or instance. The strongest open tasks remain a general source-construction benchmark, more optimized original-loop baselines, and integration with the separate Appendix D recursion. The main report gives all fixture parameters, memory tables, diagnostic counts, implementation limitations, and independent static-audit findings.

10 Contribution ledger: attribution and strength of evidence

Table 5 gives a compact answer to what is new relative to the source, what is proved, and what remains outside the demonstrated result. Section numbers in its second column refer to the full Ram report.

Table 5: Attribution and evidence, without conflating local gains with a new general complexity bound.

Item	Location	Established result	Limitation or attribution
Boolean quotient and transported products	Sections 3, 5–7	Exact equivalence and explicit observation rules.	Classical algebra and source rederivations; not a new speed bound.
Aggregated Boolean jets	Theorem 8.1; Corollary 8.2	Fewer restricted calls, with exact local generating-state implementation.	Additional coefficient cost; finite matched gains now measured; no general hard-instance guarantee or better exponent.
Matched finite implementation	Section 16	All 21 counts and full coefficient arrays agree; aggregate medians are lower on all 18 positive-order cases tested.	Small input-specific layouts, noisy timings, and potentially larger kernel storage; not a general-construction benchmark.

Item	Location	Established result	Limitation or attribution
Weighted-support extension	Theorem 9.1	Exact smaller support family when higher correction orders are certified.	No improved general source-compatible design is constructed.
Redundant-row removal	Proposition 10.1	Same fibers and counts with a spanning subset of rows.	Standard exact linear algebra; may be a no-op; rank does not bound state count.
Treewidth and rank formulations	Section 10	Exact structural algorithms or reference identities.	Restricted-input assumptions or exponential matrix dimension remain.
Actual Hermite-weight precision	Theorem 13.1	Coordinatewise modular certificate and nodewise budgets; sharp for fixed weights and independent errors.	Not globally optimal among all reconstructions or error models.
Finite precision savings	Section 13.2	Strict reductions from 64 to 7, 144 to 33, and 96 to 7 extra bits in the stated grids.	Local configurations only; not full recursive benchmarks.
Arithmetic and truncation safeguards	Sections 6, 11–15; Appendix A	Conditions under which finite operations retain the exact target.	Clarification and negative controls, not repairs to demonstrated source errors.
Quaternionic/PDE reinterpretations	Sections 6.3, 14	Precise limits on transferring continuous or noncommutative theory.	No auxiliary PDE or hypercomplex SAT speedup demonstrated.

10.1 Advantages that can be defended now

The Ram organization explicitly separates the state, coefficient algebra, product, observation functional, and exactness certificate. This makes it easier to see where information is preserved and where an apparently cheap step would discard necessary contributions. The generating-state construction is a concrete example of useful redesign guided by that organization, not merely a new name for the source formula.

The precision rule is similarly target-sensitive: it certifies the quantity actually required instead of imposing unnecessarily pessimistic precision on every input. Weighted order budgets exploit stronger structure when it is available. Exact row selection removes redundant information without changing the counting problem. These are meaningful local improvements even though their integration into the fastest source route remains unfinished.

10.2 Costs and limitations of the Ram approach

The matched benchmarks now measure gains over the chosen original-loop implementation, but do not establish an exclusive advantage unavailable to conventional generating-function algebra. All these scalar arguments can also be written in standard generating-function, quotient-ring, or Hermite-interpolation language. The report does not show that Ram terminology confers an exclusive mathematical capability. The extra generating coordinate can increase runtime constants and bit work; exact recovery weights cost time to construct; storing all reachable states can lose the polynomial-space property; hypercomplex representations can complicate a purely rational calculation.

A fair performance comparison must use the source's structured, streamed algorithm on matched inputs, not a naive dense matrix routine or brute force as the only baseline. The source's difficult incidence design and recursive arithmetic invariants remain indispensable.

11 Open questions and proposed next steps

11.1 Integration and practical performance

Proved now: the aggregated identity and its restricted-call bound. **Now implemented:** checked finite layouts, including five node-varying designs. **Still needed:** implementation within the general Section 4 construction of the source and comparison with stronger optimized baselines. Compare identical instances under the pair loop and aggregated loop, tracking ring operations, coefficient bit lengths, peak live memory, and elapsed time. A successful result would identify parameter ranges where saved calls outweigh the larger coefficient state. The matched finite comparison is reported in revision 2; a general-construction comparison is not.

11.2 Higher-order designs and a stronger exponent

Proved now: weighted aggregation under certified vanishing orders. **Still needed:** construct those orders while controlling deformation degree, summary ranges, incidence count, and coefficient heights. In the earlier route the exponential cost contains three interacting contributions, schematically

$$O\left(n\left[\frac{R}{Q}\log\frac{2Q}{R}+\tau\log(2D+1)+\frac{1}{r}\right]\right).$$

A reduced subset family is not enough if another term grows. Improving the sharper Appendix D bound further requires analysis of that distinct recursion. The report supplies neither construction nor complete new complexity inequality establishing such an improvement.

11.3 Recursive precision savings at scale

Proved now: exact target weights give valid precision budgets and strict finite savings. **Still needed:** integrate them while retaining immutable creation-time lifts and all divisibility invariants; measure nontrivial recursive instances; and establish better uniform valuation bounds if an asymptotic gain is claimed. Finite cancellations are encouraging evidence for local optimization, not proof of a better worst-case recurrence.

11.4 Finite-domain and weighted-count generalizations

Exact finite-alphabet algebras and signed weighted sums are natural extensions of the reformulation. But the source already allows commutative $\mathbb{Q}$-algebras in its earlier stage, so this observation alone is not a new theorem. A genuinely extended complexity result would need compatible designs, bounded domain encodings, factor degrees and scopes, coefficient heights, and a reconstruction modulus sufficient for the weighted answer. The $n + 1$-bit bound for an ordinary Boolean count does not automatically apply to arbitrary rational or signed weights.

11.5 Formal verification, historical novelty, and lower bounds

The finite-sum aggregation proofs and the valuation certificate are suitable for formalization. A complete deliverable would connect the identities to coefficient arrays, exact arithmetic, streamed storage, and the source interfaces. An abstract identity alone would not certify the implementation. That formalization has not been carried out in this report.

A more extensive prior-art search is also required before claiming that the refinements are historically novel. Finally, no lower bound proves the source exponent optimal. Examples with many states or ill-conditioned reconstruction show limitations of particular methods, not impossibility of every faster SAT algorithm. Both a stronger algorithm and an optimality theorem remain separate unresolved tasks.

12 Conclusion

The clearest positive result is an exact replacement of a repeated subset-pair enumeration by a subset enumeration with a small generating state. This is accompanied by a conditional weighted-support extension, exact summary-row compression, and precision certificates that improve conservative allowances in verified finite examples. The full report provides proofs and exact tests, not merely a speculative analogy.

Equally important, it does not confuse those results with a new solution of the global complexity problem. The source's principal construction and sharp recursive bounds are retained. Revision 2 measures finite matched pipeline speedups, not a stronger general SAT exponent, a performance advantage on arbitrary instances, or an independent complete refutation of ETH. The benefits are exact local refinements, clearer accounting of information and arithmetic, and a concrete research program for testing whether those local gains survive integration. That is the appropriate interpretation of the present contribution relative to the OpenAI paper.

References

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[4] OpenAI. *SubexponentialSAT solution interfaces and comparator challenge*. Snapshot inspected in [1]: `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/lean/OAI/Algorithms/SubexponentialSAT. This summary inherits the bounded inspection described in the main report; no independent Lean build is claimed.